

Jaisidh Singh

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Education

2024–present **MSc in Machine Learning**, *Eberhard-Karls Universität Tübingen*, Tübingen.
(expected 2027)

- German GPA: 1.52/5.0
- Thesis: scaling behaviour of hybrid-attention LLMs.
- Publications: 1× EMNLP Main Conference, 2× ICLR Workshops, 1× NeurIPS Workshops.

2020–2024 **B.Tech in AI & Data Science**, *Indian Institute of Technology*, Jodhpur.

- Thesis: negation-aware compositional reasoning in vision-language models.

Awards & fellowships: Konrad Zuse School (ELIZA) Fellowship for Master's Students in Germany 2024–2026.

Work/Research Experience

Oct 2026 – **AI Research Intern at Aleph Alpha**, *With Dr. Yasser Jadidi & Dr. Leonard Salewski*.
present

- Working on data generation for SFT.

Apr 2026 – **Thesis Researcher at OpenEuroLLM**, *With Dr. Aaron Klein & Dr. Antonio Orvieto*.
present

- Improving data efficiency ($\approx 30\%$)** in LLM pretraining by scaling hybrid-attention architectures.
- Contributed to novel linear attention architecture research** via language modeling experiments.
Published: NeurIPS 2026 NeuReps Workshop.

Oct 2025 – **SWE Graduate Assistant at Tübingen AI Center**, *With Prof. Peter Gehler*.
May 2026

- Designed & deployed to production** real-time interactive computer vision systems for artistic stylization of user images at Stadtmuseum Tübingen using semantic segmentation and pose estimation based warping.
- Developed & **delivered a hands-on workshop** on agent orchestration at **GCPR 2025**.

Oct 2025 – **Graduate Researcher at ELLIS Institute Tübingen**, *With Dr. Antonio Orvieto*.
Jan 2026

- Investigated grokking dynamics in transformers**, showing CE+AdamW does not admit kernel regime & layer-norm modulates grokking speed. **Published: ICLR 2026 Trustworthy AI Workshop.**

Jan 2025 – **ML Research Assistant at ELLIS Institute Tübingen**, *With Dr. Antonio Orvieto*.
May 2025

- Reduced VLM training cost by 90%** for modality stitching by using a hypernetwork-based framework.
Published: EMNLP 2025 Main Conference.

Winter 2022 & 2023 **Undergraduate Researcher at Trusted AI Lab IITJ**, *With Prof. Mayank Vatsa, Prof. Richa Singh*.

- Designed a compositional negation benchmark & fine-tuned CLIP on custom data, **achieving SOTA accuracy on negation understanding**. **Published: WACV 2025 (60+ citations).**
- Systematically analyzed StyleGAN2 latent space to profile identity leakage. **Published: WACV 2024.**

Summer 2022 & 2023 **Research Engineer Intern at Bosch Research India**, *With Dr. Amit A. Kale & Sonam Singh*.

- Developed an interpretable failure-discovery system** for semantic segmentation models for industry-scale autonomous driving data **achieving 95% precision**. **Technical report: arXiv:2411.10845.**
- Created multi-modal image retrieval pipelines** analyzing weather effects on AV cameras.

Selected Publications

2025 **(Almost) Free Modality Stitching of Foundation Models**, *EMNLP 2025, ICLR-W 2025*, Jaisidh Singh, Diganta Misra, Boris Knyazev, Antonio Orvieto.

2025 **Learning the Power of "No": Foundation Models with Negations**, *WACV 2025*, Jaisidh Singh*, Ishaan Shrivastava*, Mayank Vatsa, Richa Singh, Aparna Bharati.

(Refer to my Google Scholar profile for a complete list of publications. * indicates equal contribution.)

Technical Skills

- Machine learning:** PyTorch, JAX, FLAX, Haiku, Optax, Huggingface
- Distributed training & HPC:** Slurm, Megatron-LM, Sharding/distributed training
- Agentic frameworks:** dsPy, n8n, LangGraph, LangChain
- Programming languages:** Python, C++, Bash, JavaScript, Dart